Unification of Gravitational and Electromagnetic Forces via Complex Charge: Perihelion Advance, Galactic Rotation, Relativistic Beam Deflection, and a Local Velocity Measurement Prediction

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https://github.com/irxyzcameron6D/cameron-unified-framework

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Abstract

We present a framework in which all four fundamental forces arise as projections of the single complex force law $F = Q_1Q_2/r^2$, where the complex charge $Q = q + im\sqrt{G}$ unifies electromagnetic and gravitational coupling. Newton's law follows exactly from $i^2 = -1$ with no free parameters. The nuclear van der Waals force is confirmed at 8–14σ by Monte Carlo simulation. The circumgalactic medium shell mass mechanism reproduces the Large Magellanic Cloud orbital velocity to 0.3% using only baryonic matter. A velocity-dependent imaginary sector coupling $M_{eff} = M(1 + v^2/c^2)$ derives the Mercury perihelion advance from three contributions in a strict 3:2:1 ratio summing to 43.00 arcseconds per century — a structural signature absent from General Relativity. The same coupling predicts that relativistic neutral particle beams deflected by the Sun follow $\alpha = 2GM(1+\beta^2)\sqrt{(1-\beta^2)} / (bc^2\beta^2)$, approaching zero as $\beta \to 1$ for massive particles before jumping to the Eddington value for photons — a hard discontinuity absent from General Relativity. A ship moving at speed β through the interstellar medium experiences a fore/aft temperature asymmetry $\beta = (R-1)/(R+1)$ where $R = T_{forward}/T_{aft}$, providing a local absolute velocity measurement without observing the CMB dipole. All numerical failures of the framework are documented alongside the successes.

Keywords: gravitational unification, complex charge, van der Waals gravity, galactic rotation, perihelion advance, gravitational lensing, relativistic beam deflection, interstellar medium

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I. Introduction

The Standard Model and General Relativity rest on incompatible mathematical foundations. The vacuum energy discrepancy of 10^{120}, the absence of a detected dark matter particle after decades of search, and the Hubble tension all suggest the current framework is incomplete. In three published papers [1,2,3], Cameron proposed that gravity is the imaginary sector of a unified complex force law, that missing galactic mass is circumgalactic medium gas excluded from standard budgets, and that cosmological redshift has a gravitational component beyond standard Doppler. This paper collects and extends those results, presenting three new derivations not in the published papers: the Mercury perihelion advance from first principles (Section IV), a relativistic beam deflection prediction that distinguishes this framework from General Relativity (Section V), and a local velocity measurement from hull temperature asymmetry (Section VI). All open problems are documented in Section VII.

II. The Complex Charge and Unified Force Law

We define the complex charge of a particle as (in Gaussian CGS units):

$$Q = q + im\sqrt{G} \tag{1}$$

where q is electric charge, m is invariant rest mass, G is Newton's constant. The unified force law $F = Q_1Q_2/r^2$ expands as:

$$Q_1Q_2 = q_1q_2 - Gm_1m_2 + i(q_1m_2+q_2m_1)\sqrt{G} \tag{2}$$

The three terms are electromagnetism (real×real), gravity (imaginary×imaginary, attractive from $i^2=-1$), and nuclear van der Waals (cross term). For neutral hydrogen $F = -G(m_p+m_e)^2/r^2$ exactly, with no parameters. The neutron at phase angle $\phi = \pi/2$ (pure imaginary charge) produces nuclear van der Waals binding confirmed at 8–14σ by Monte Carlo simulation [1].

III. Galactic Rotation and the LMC Orbital Test

For CGM density $\rho \propto R^{-2}$, enclosed mass $M(<R) \propto R$, giving flat rotation curves without dark matter. The Large Magellanic Cloud at R = 50 kpc orbits outside the galactic disc; the Shell Theorem applies exactly. Required enclosed mass:

$$M_{MW}(<50\text{ kpc}) = V_{tan}^2 \times R_{LMC} / G \tag{3}$$

Cameron prediction: V = 280.2 km/s. Observed (Gaia+HST [4]): V_{tan} = 281 ± 41 km/s. **Agreement: 0.3%**, using only baryonic matter. Current X-ray surveys account for ~68% of the required CGM mass. The remaining 32% is a testable prediction for Athena and LynX.

IV. Perihelion Advance from First Principles

As a body moves through a gravitational field, its velocity couples increasingly to the imaginary sector. The effective gravitational mass seen by a particle at speed $v = \beta c$:

$$M_{eff} = M(1 + v^2/c^2) = M(1 + \beta^2) \tag{4}$$

At $v = 0$: $M_{eff} = M$ (Newton). At $v = c$: $M_{eff} = 2M$ (the Eddington factor — confirmed 1919). Both the perihelion advance and the Eddington lensing follow from equation (4).

A. The 3:2:1 Structure

The total advance $\delta\phi = 6\pi GM/(c^2p)$ arises from three contributions in the strict ratio 3:2:1:

Contribution	Formula	arcsec/century	Fraction
PMV kinematics (Binet)	$3\pi GM/(c^2p)$	21.50	3/6
Imaginary sector coupling	$2\pi GM/(c^2p)$	14.33	2/6
Gravitomagnetic	$\pi GM/(c^2p)$	7.17	1/6
Total	$6\pi GM/(c^2p)$	43.00	6/6

Table 1. Three contributions to Mercury perihelion advance. The 3:2:1 ratio is a specific Cameron structural signature.

The 3:2:1 ratio distinguishes this derivation from any approach attributing the full result to a single cause. Verified for all planets simultaneously (Table 2).

Planet	a (AU)	e	Predicted (arcsec/cy)	Observed
Mercury	0.3871	0.2056	43.00	43.0 ± 0.5
Venus	0.7233	0.0068	8.63	8.6 ± 4.7
Earth	1.0000	0.0167	3.84	5.0 ± 1.2
Mars	1.5237	0.0934	1.35	1.35 ± 0.65

Table 2. Planetary perihelion advance. All predictions match GR and observation.

V. Relativistic Beam Deflection — A Testable Prediction

Neutral atoms or neutrons (no net electric charge) deflected by the Sun isolate the Cameron coupling from electromagnetic confounding effects. The Cameron framework couples gravity to the imaginary charge $\mathrm{Im}(Q) = m\sqrt{G}$, where m is the *invariant rest mass* — Interpretation B of special relativity consistently applied to both kinematics and gravitational coupling. General Relativity couples to the stress-energy tensor (which includes γm), giving a smooth transition to the Eddington limit. The Cameron deflection formula:

$$\alpha_{\text{Cameron}} = 2GM(1+\beta^2)\sqrt{(1-\beta^2)} \; / \; (bc^2\beta^2) \tag{5}$$

$\beta = v/c$	Newton (arcsec)	GR (arcsec)	Cameron (arcsec)	Cameron/GR
0.10	87.59	88.46	88.02	0.995
0.50	3.50	4.38	3.79	0.866
0.786	1.42	2.29	1.42	0.618
0.90	1.08	1.96	0.85	0.436

0.99	0.89	1.77	0.25	0.141
1.000 (massive→c)	∞	1.752	0.000	0
photon	∞	1.752	1.752	1.000

Table 3. Predicted deflection angles for neutral beams grazing the Sun ($b = R_{Sun}$ = 696,000 km). Eddington photon value: 1.752 arcseconds.

Three distinguishing features: (i) At $\beta = 0.786$ Cameron equals Newton while GR predicts 61% more. (ii) At $\beta = 0.99$ Cameron gives 0.25 arcsec versus GR 1.77 arcsec — a factor of seven. (iii) A hard discontinuity at v = c: massive particle deflection → 0, photon deflection = 1.752 arcsec (Eddington). GR predicts smooth transition. An experiment at $\beta = 0.9$ with 20% precision would be decisive. This prediction is connected to the Schwarzschild result that coordinate velocity of infalling matter peaks at $r = 3r_s$ then decreases to zero at the horizon — both are coordinate observables vanishing as the null geodesic boundary ($d^2 = 0$) is approached.

VI. Local Velocity Measurement — The Hull Temperature Speedometer

A ship moving at speed β through the ISM/CMB medium experiences a real physical asymmetry — not a coordinate effect. The forward hull intercepts CMB photons and ISM atoms blueshifted by the relativistic Doppler factor; the aft hull is in near-vacuum with redshifted radiation:

$$T_{fwd} = T_0 \times \sqrt{(1+\beta)/(1-\beta)} \quad T_{aft} = T_0 \times \sqrt{(1-\beta)/(1+\beta)} \tag{6}$$

The ratio $R = T_{fwd}/T_{aft} = (1+\beta)/(1-\beta)$. Solving for β:

$$\beta = (R - 1) / (R + 1) \text{ where } R = T_{\text{forward hull}} / T_{\text{aft hull}} \tag{7}$$

This gives velocity relative to the CMB rest frame from a *local* measurement — no need to observe the CMB dipole. At $\beta = 0.99$: R = 199; at $\beta = 0.999$: R = 1999. The method is more sensitive at high β than the standard CMB dipole. The forward hull pressure diverges as $\beta \to 1$ while aft pressure → 0, physically crushing the ship in the direction of motion. This physical compression is distinct from Lorentz contraction (a coordinate effect producing no physical stress). The Penrose-Terrell theorem (1959) is exact for motion through vacuum. In a physical medium, real asymmetric forces physically compress the ship — superimposed on, and distinct from, the optical rotation effect. Both are present simultaneously in a medium.

VII. Open Problems and Documented Failures

The following are unresolved. They are documented rather than omitted.

Gap 1 — Full orbital average. Local (3,12) mean gives -5.77×10^{-64} N/pair (factor 3 from Newton). Full orbit average: -3.62×10^{-69} N (factor 50,000 too small). Selection mechanism not yet derived.

Gap 2 — R^{-2} scaling. Dipole-dipole forces scale as R^{-4}. The orbit averaging that yields R^{-2} for flat rotation curves is demonstrated numerically but not analytically derived.

Gap 3 — Relativistic γ in Monte Carlo. Quarks move at ~0.99c. The γ correction is not yet implemented. Expected signal increase: ~49×.

Gap 4 — G from first principles. Master equation gives G within factor ~9. Requires Gap 3 first.

Gap 5 — Outer disc rotation. Two-component model gives V ~ 139 km/s at 50 kly vs observed ~221 km/s. Multi-phase CGM mass census incomplete.

VIII. Conclusion

The complex charge framework $Q = q + im\sqrt{G}$ unifies electromagnetism and gravity with no free parameters. The LMC orbital test agrees with observation at 0.3% using only baryonic matter. The Mercury perihelion advance is derived from three Cameron contributions in the ratio 3:2:1, summing to 43.00 arcseconds per century, matching GR and observation for all planets simultaneously.

The relativistic beam deflection prediction — massive particle deflection $\to 0$ as $v \to c$, photons retain the full Eddington value — distinguishes this framework from General Relativity by a factor of seven at $\beta = 0.99$. An experiment at $\beta = 0.9$ with 20% precision is decisive.

The hull temperature speedometer $\beta = (R-1)/(R+1)$ provides a local absolute velocity measurement without observing the CMB dipole, sensitive at high β where other methods are imprecise.

All Monte Carlo scripts, derivation files, and documented failures are available at [https://github.com/irxyzcameron6D/cameron-unified-framework]. Reproduction of all numerical results is encouraged.

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References

[1] D. Cameron, "Is the current estimation of the ratio of dark matter to visible matter the result of a simple miscalculation", Physics Essays 25, 306 (2012).

[2] D. Cameron, "Is the force of gravity a manifestation of the electric force?", Physics Essays 28, (2015)

[3] D. Cameron, "Donald Cameron, Does the cosmological red shift have a gravitational component?", Physics Essays 31 (2018).

[4] [Kallivayalil, N., van der Marel, R. P., Alcock, C., et al. (2006). The Proper Motion of the Large Magellanic Cloud Using HST. The Astrophysical Journal, 638(2), 772-785.

[5] J. Tumlinson et al., "The COS-Halos Survey", Astrophys. J. 777, 59 (2013).

[6] Zhang, Y., et al. (2024), "The hot circumgalactic medium in the eROSITA All-Sky Survey - II. Scaling relations between X-ray luminosity and galaxies' mass," Astronomy & Astrophysics, 690, A268.

[7] B. P. Abbott et al. (LIGO/Virgo), "GW170817", Phys. Rev. Lett. 119, 161101 (2017).

[8] M. Ahmadi et al. (ALPHA Collaboration), "Observation of the effect of gravity on the motion of antimatter", Nature 621, 716 (2023).

[9] R. Penrose, "The apparent shape of a relativistically moving sphere", Proc. Cambridge Phil. Soc. 55, 137 (1959).

[10] J. Terrell, "Invisibility of the Lorentz contraction", Phys. Rev. 116, 1041 (1959).

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